TIPA Korea Technology and Information Promotion Agency for SMEs

(max 10 pages)

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

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2. Technology Overview

Title of Proposed Technology	Cell-level digital twin for anode-free lithium-metal battery (AF-LMB) pouch cells: multi-physics model and safety prediction toolbox			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display ( ), Secondary Battery (√), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio ( ), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), AI (√), Next-gen Communication ( ), Advanced Robotics & Manufacturing ( ), Quantum ( )			
Industrial Technology Classification [1)]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Electrical & Electronics		Electrical & Electronics	
	Medium Category	Small Category	Medium Category	Small Category
	Instrumentation	Safety monitoring/diagnosis measurement and control instruments	Batteries	Secondary Batteries
Concept of Proposed Technology	Develop a cell-level digital twin for anode-free lithium-metal battery (AF-LMB) pouch cells to reproduce and predict cell behavior under realistic operating, environmental, and manufacturing conditions. [1][2]			

	Core functions: • High-fidelity multi-physics model including electrochemical dynamics (Li transport and reaction kinetics), Li plating and stripping behavior, SEI growth and interfacial degradation, thermal and mechanical coupling, and tab and edge current distribution. • Degradation and safety prediction toolbox to evaluate Li plating onset, void formation, hotspot development, cycle-life degradation pathways, and safety thresholds such as allowable current and thermal limits. • Digital twin engine for performance prediction under application mission profiles, formation optimization (pressure, temperature, C-rate), and manufacturing variation analysis (porosity, loading, thickness), including sensitivity mapping for QC and process control. • Diagnostic and control-oriented outputs such as safe operating area maps (SOC, temperature, C-rate), indicators for remaining useful life (RUL) and state of health (SoH), and early anomaly detection features based on voltage, thermal, and impedance signatures.
Rationale for Joint R&D Project	Anode-free lithium-metal battery (AF-LMB) cells have specific failure and safety risks including Li plating, void formation, local hotspots, tab-edge current concentration, and sensitivity to formation conditions. The Korean SME partner seeks to accelerate development cycles and strengthen safety and reliability by using a computational digital twin to reduce reliance on long experimental iteration, to define safe operating envelopes for future BMS and pack-level control, and to support manufacturing parameter optimization. Collaboration with TNO is requested to apply TNO's multi-physics modeling, diagnostic frameworks, and system-level analysis capabilities to build and validate this AF-LMB digital twin platform together.
Key Words	1. Anode-free lithium-metal battery (AF-LMB) 2. Cell-level digital twin 3. Multi-physics electrochemical-thermal-mechanical modeling 4. Degradation and safety prediction 5. Formation optimization and safe operating envelope

3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	Start of the Project	Level (3)	End of the Project	Level (6)
R&D Objectives	• Develop a physics-based multi-physics AF-LMB pouch-cell model covering electrochemistry, Li plating/stripping, SEI growth, and thermal-mechanical coupling, including current distribution effects at tabs and edges. [5] • Create a degradation and safety prediction toolbox to map Li plating onset, void and hotspot evolution, cycle-life degradation pathways, and safety thresholds. • Implement a cell-level digital twin engine enabling virtual evaluation of mission profiles, formation optimization, and manufacturing variation and sensitivity analysis for QC. [3] • Generate control-oriented outputs: safe operating area maps (SOC-temperature-C-rate), SoH/RUL indicators, and early anomaly detection features for future BMS and pack engineering.			
R&D Target Specifications	Target deliverables and quantitative coverage (based on data available from the Korean SME partner): • Cell formats covered for model calibration/validation: AF-LMB pouch cells in the 17 Ah and 20 Ah class. • Operating temperature envelope represented in the model datasets: discharge down to -35°C and charge from +10°C (as reported for the SME partner's products). • Current-rate coverage represented in the model datasets: continuous discharge up to 8C to 10C and pulse discharge up to 20C (10 seconds), aligned with the SME partner's published product limits. Outputs: safe operating area map spanning SOC 0% to 100%, temperature, and C-rate; formation optimization maps (pressure, temperature, C-rate); and manufacturing-variation sensitivity maps (porosity, loading, thickness). Note: numeric prediction accuracy targets (for example voltage error bands) should be agreed after the joint definition of validation datasets.			
R&D Plan	Draft work plan (to be aligned with the official 2026 TIPA RFP schedule and project duration): **Work Package 1: Requirements and datasets (Months 0 to 3)** • Define use cases and mission profiles (UAV, robotics, high-power tools) and the validation matrix. • Collect and curate cell design, manufacturing, and test datasets from the Korean SME partner. **Work Package 2: Multi-physics cell model development (Months 3 to 9)**			

	• Build coupled electrochemical, thermal, and mechanical model including Li plating/stripping and SEI growth. • Incorporate tab and edge current distribution. **Work Package 3: Parameterization and validation (Months 6 to 15)** • Parameter identification using experimental datasets. • Validate model behavior across temperature and C-rate ranges considering 2D temperature and pressure heterogeneity. **Work Package 4: Degradation and safety prediction (Months 9 to 18)** • Implement algorithms for plating onset, void formation, hotspot development, and safety thresholds. **Work Package 5: Digital twin engine and diagnostic outputs (Months 12 to 22)** • Build the digital twin workflow for mission profile simulation, formation optimization, and manufacturing variation analysis. • Produce safe operating maps and SoH/RUL and anomaly indicators. **Work Package 6: Demonstration and knowledge transfer (Months 18 to end)** • Demonstrate the digital twin on representative cell datasets and application scenarios. • Deliver documentation, training, and a handover package supporting internal use by the Korean SME partner.

4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	The Research Institute (TNO) is expected to: • Develop physics-based multi-physics AF-LMB cell models covering electrochemical behavior, lithium plating/stripping, SEI growth, and electro-thermal-mechanical coupling. • Implement degradation and safety prediction algorithms, including lithium plating onset, void formation, hotspot development, and safety threshold identification. • Build and deliver a cell-level digital twin engine enabling virtual evaluation of mission profiles, formation optimization, and manufacturing-variation sensitivity analysis. • Generate control- and BMS-oriented outputs, including safe operating area maps (SOC–temperature–C-rate), state-of-health (SoH), remaining useful life (RUL), and early anomaly detection indicators. • Lead model parameterization, uncertainty quantification, and validation using experimental datasets and established battery modeling methodologies. • Utilize battery laboratory and test infrastructure for calibration and verification, including electro-thermal characterization, ageing analysis, and algorithm validation (MiL/HiL where applicable). • Provide documentation, training, and knowledge transfer to enable the Korean SME partner to apply, maintain, and extend the digital twin framework during and after the project.
Expected R&R of Korean SME Partner	Expected contributions of the Korean SME partner: • Provide AF-LMB pouch-cell designs, manufacturing process information, and representative production-variation parameters (for example porosity, loading, thickness) needed for modeling. • Produce and supply cell samples and perform laboratory tests for calibration and validation across temperature and C-rate conditions. • Provide existing test and validation protocols and datasets for performance, safety, and reliability. • Define application mission profiles and integration needs for future pack and BMS use. • Evaluate digital twin outputs and support demonstration on relevant application scenarios.

5. Expected Outcomes/Impact

Expected Outcomes and Impact	Expected outcomes: • A validated, AF-LMB specific cell-level digital twin platform that supports virtual evaluation of performance, degradation, and safety under operating, environmental, and manufacturing conditions. • Reduced experimental iteration by enabling simulation-guided formation optimization and manufacturing parameter tuning. • Safety and reliability evidence packages, including safe operating envelopes and early warning indicators, supporting customer qualification and certification preparation. Expected impact: • Faster development cycles and more robust deployment of AF-LMB cells for demanding applications such as defense UAVs, robotics, and high-power tools. • Strengthened competitiveness of the Korean SME partner through improved design-for-safety and design-for-manufacture capabilities enabled by the joint TNO collaboration.

6. References

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(3) Rath, S., Medina, R., Wilkins, S. Real-time optimal charging strategy for a fleet of electric vehicles minimising battery degradation. TNO publication, 2023. Accessed 2025-12-16. https://publications.tno.nl/publication/34641555/EemswK/rath-2023-real.pdf

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(5) Lodge, A., Medina, R., Hoekstra, F. S. J., Wilkins, S. Embedded Physics-based Electrochemical Models for Real-time Algorithms. TU/e Research Portal (conference poster), published 22 May 2025. Accessed 2025-12-16. https://research.tue.nl/en/publications/embedded-physics-based-electrochemical-models-for-real-time-algor/

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(7) BPX Standard. July 2025 release page (BPX physics-based battery modelling standard). 3 Jul 2025. Accessed 2025-12-16. https://bpxstandard.com/2025/07/

☐ Terms:

1) Industrial Technology Classification

· Classification system according to Industrial Technology Classification Table (**Appendix 1**, based on Article 16 of the Common Operational Guidelines for the Industrial Technology Innovation Project in Korea).

2) Technology Readiness Level (TRL)

Description		
·A consistent and objective indicator of the technological maturity of key element technologies		
	Basic Research Stage (1-2)	1st Stage: Basic theories and experiments 2nd Stage: Proof of Concept, such as practical ideas or patents
	Experimental Stage (3-4)	3rd Stage: Verification of basic performance at a laboratory scale 4th Stage: Evaluation of key performance of materials, components, and systems at a laboratory scale
	Prototype Stage (5-6)	5th Stage: Production and performance evaluation of prototypes of finalized materials, components, and systems 6th Stage: Production and performance evaluation of prototypes at a pilot scale
	Productization Stage (7-8)	7th Stage: Reliability testing and evaluation by potential customers 8th Stage: Prototype certification and standardization
	Commercialization Stage (9)	9th Stage: Commercialization